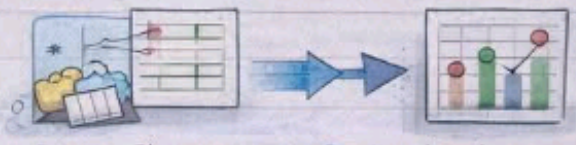
< PANDAS IN PYTHON >

What is Pandas?

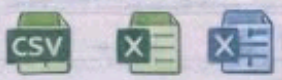
- ✓ Full form: Panel Data
- ✓ Pandas is an open-source Python library for data manipulation and analysis.

Why Use Pandas?

- ✓ Data Cleaning
- ✓ Data Transformation
- ✓ Data Analysis
- ✓ Supports CSV, Excel, SQL, etc.



• Data Cleaning • Data Analysis



• SQL, etc.

Pandas vs NumPy

Pandas		NumPy
✓ DataFrame & Series	VS	✓ N-dmensional array
✓ Multi-dimensional		✓ Numerical computations

Applications:

- ✓ Data Analysis
- ✓ Machine Learning
- ✓ Time Series Analysis



```
import pandas as pd
```

- Data sources Like user

DataFrame

Name	Age	City
Alice	25	New York
Bob	30	Los Angeles
Charlie	35	Chicago

< PANDAS - INSTALLATION & IMPORT >

Installing Pandas

✓ Pandas ko install karne ke liye command:

```
pip install pandas
```

Importing Pandas

```
import pandas as pd
```

pd. → short alias (easy & standard)

Why Pandas is Imported as pd?

✓ Code short hota hai
✓ Industry standard practice
✓ Easy to read & write

☆IMPORTANT NOTES☆

✓ Pandas is built on NumPy
✓ Mainly used for data analysis
✓ Works with Series & DataFrame

First Pandas Example

```
import pandas as pd
data = [10, 20, 30]
s = pd.Series(data)
print(s)
```

Output

```
0    10
1    20
2    30
dtype: int64
```

☆IMPORTANT NOTES☆

✓ Pandas is built on **NumPy**
✓ Mainly used for data analysis
✓ Works with Series & DataFrame



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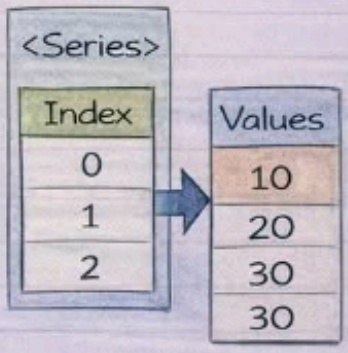


Pandas is a Python library used for data manipulation and analysis.

< PANDAS SERIES >

What is a Series?

- ✓ A one-dimensional labeled array in Pandas.
- ✓ Stores data of any type (Integers, strings, etc.)
- ✓ Similar to a column in a DataFrame.

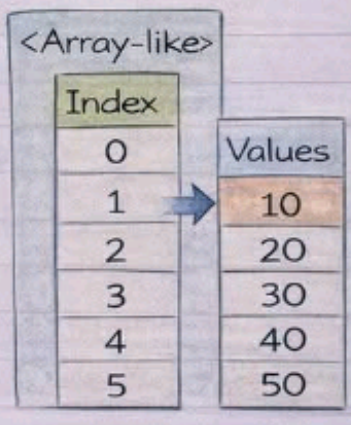


Creating a Series

- ✓ Create from a list
- ✓ Import pandas as pd
- ✓ Use the pd.Series() function

Index & Values

- ✓ Index are like labeled address for each value.
- ✓ Default index starts from 0
- ✓ Values are the data stored at each label.



Example

```
import pandas as pd
data = [10, 20, 30, 40, 50]
s = pd.Series(data)
print(s)
```

Output

```
0    10
1    20
2    30
3    40
4    50
dtype: int64
```

Pandas is a Python library used for data manipulation and analysis.

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< PANDAS DATAFRAME >

What is a DataFrame?

- ✓ A two-dimensional labeled data structure.
- ✓ Stores data in a table like format.
- ✓ Think of it as a table (rows and columns).

<Index>

Name	Age	City
Alice	25	New York
Bob	30	Los Angeles
Charlie	35	Chicago

<Columns>

Creating a DataFrame

- ✓ From a dictionary
- ✓ From a list of lists
- ✓ From a NumPy array

```
import pandas as pd
data = [
 'Name'= ['Alice', 'Bob', 'Charlie'],
 'Age' = [25, 30, 35],
 'City'= ['New York', 'Los Angeles', ' Chicago']
df = pd.DataFrame(data)
```

Rows & Columns

- ✓ Rows have an index starting from 0
- ✓ Columns have labels
- ✓ Shape:
  - Represents DataFrame dimensions
  - Tuple of (rows, columns)

<Rows>

	Name	Age	City
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Charlie	35	Chicago

<Columns>

```
import pandas as pd
data = [10, 20, 30, 40, 50]
s = pd.Series (data)
print(s)
```

Name	Age	City
Alice	25	New York
Bob	30	Los Angeles
Charlie	35	Chicago

(3, 3)

< READING & WRITING DATA >

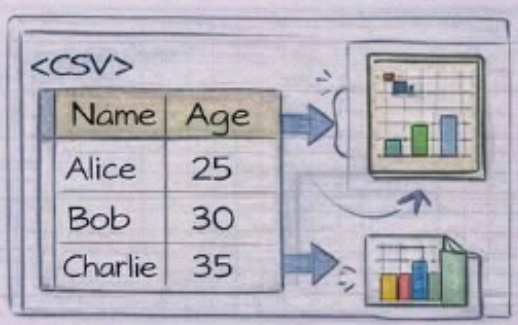
Reading Data

✓ Reading from CSV:

✓ Reads data from a CSV file

✓ Creates a DataFrame from the data

```
df = pd.read_csv(data.csv')
```

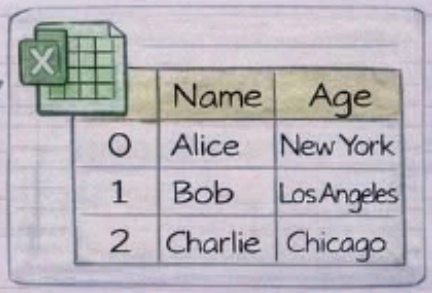


Reading from Excel :

✓ Reads data from an Excel file

✓ Creates a DataFrame from the data

```
df = pd.read_excel('data.xlsx')
```



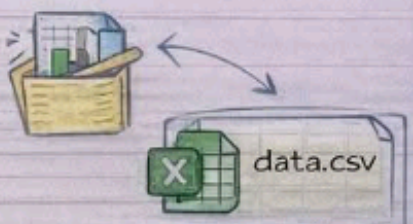
	Name	Age
0	Alice	New York
1	Bob	Los Angeles
2	Charlie	Chicago

Writing Data

✓ Writing to CSV:

✓ Writes DataFrame to a CSV file

✓ Converts to tabular text with comma separator




```
df.to_csv('newdata.csv',
          index=False)
```

Name	Age	City
0	25	New York
1	30	Los Angeles
2	35	Chicago

(3, 3)

Writing to Excel

✓ Writes DataFrame to an Excel file

✓ Converts to a spreadsheet format

```
df.to_csv('newdata.xlsx', index=False)
```

< PANDAS OPERATIONS >

head()

✓ Shows the first n rows (default = 5)

```
df.head()
```

Name	Age	City
Alice	25	New York
Bob	30	Los Angeles
Charlie	35	Chicago

First 5 rows

tail()

✓ Shows the last n rows (default = 5)

```
df.tail()
```

....	....	....	City
97	Jane	45	Seattle
98	Mike	50	Boston
99	Sara	55	Austin

Last 5 rows

shape & info()

✓ shape: Returns (#rows, #columns)

```
df.shape
```

dfdescribe()

```
df.describe()
```

✓ Displays summary statistics

```
df.shape
```
(100, 3)

✓ Displays DataFrame info

```
<<class 'panndas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 3 columns):
```

#	Column	Non-Null Count	Dtype
0	Name	100 non-null	object
1	Age	100 non-null	int64
2	City	100 non-null	object

df.describe()

✓ Displays summary statistics

```
df.describe()
```

count.	std	min	S%	max
100.0	35.7%	18.0	25.0	55.0
25.0	30.0	25.0	30.0	55.0

< PANDAS INDEXING >

Selecting Columns

✓ Access one or more columns of a DataFrame.

```
df['Name']
```

```
df["Name', 'Age']]
```

Name	Age	City
Alice	25	New York
Bob	30	Los Angeles
Charlie	35	Chicago

First 5 rows

Selecting Rows

✓ Access row by index label

✓ Use loc[] for label-based indexing

✓ Use iloc[] for position-based indexing

< With loc] >

Name	Age	City
97	45	Seattle
98	50	Boston
99	55	Austin

```
df.loc[2]
```

Last 5 rows

```
df.iloc[0]
```

< With iloc] >

Name	Age	City
Alice	25	New York

Conditional Filtering

✓ Select rows based on a condition

```
df[df['Age] > 30]
```

<Condition

Age > 30

Name	Age	City
Bob	30	Los Angeles
Charlie	35	Chicago

Multiple Conditions

✓ Combine conditions using operators

```
df[df['Age'] > 25) & (df['City'] == 'New York')]
```

Age> > 25 & New York		Age	City
Alice	25	25	New York

< PANDAS REVISION & VIVA >

Quick Definitions:

- ✓ Pandas is a Python library for data analysis.
- ✓ Series is a one-dimensional labeled array in Pandas.
- ✓ DataFrame is a two-dimensional labeled data structure.

Important Functions Recap:

`read_csv()` Reads data from a CSV file ----->

`head(), tail()` Display first / last n rows ----->

`info(), describe()` Show DataFrame info a summary stats

Series		DataFrame
Alice	25	New York
Bob	30	Los Angeles
Charile	35	Chicago

Last 5 rows

Difference: Series vs DataFrame

Series	DataFrame
✓ 1-Dimensional (1 column)	✓ 2-Dimensional (multiple columns)
✓ Stores a labeled array	✓ Stores a tabular data structure

Viva Questions:

- ✓ What is Pandas?
- ✓ Difference between Series & DataFrame?
- ✓ How to read a CSV file into a DataFrame?
- ✓ What does head(), tail(), info() & describe() do?

Summary

Age > 25 & New York

Name	Age	City
Alice	25	New York

Summary

Pandas is a powerful Python library for data manipulation and analysis.

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